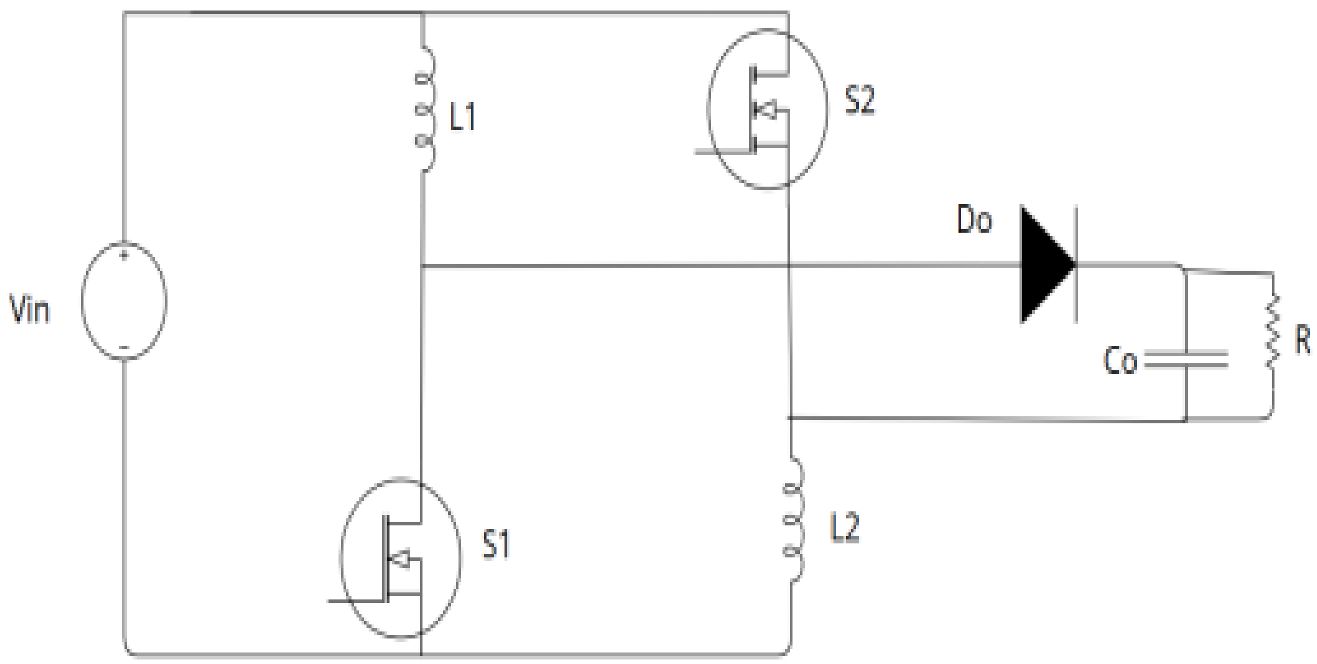
Circuit diagram



CALCULATIONS

Problem Statement: Vin=48V, Vo = 200 V, Po = 200 W, ΔIL=0.25×Io, ΔVo=0.05×Vo, fs = 50 kHz

Design Calculations

1. Duty Ratio Calculation

The voltage gain of the proposed boost converter is given by

$$\frac{V_o}{V_{in}} = \frac{1+D}{1-D}$$

Given,

- Input Voltage, $V_{in} = 48$ V
- Output Voltage, $V_o = 200$ V

$$\frac{200}{48} = \frac{1+D}{1-D}$$
$$4.167 = \frac{1+D}{1-D}$$

Cross multiplying,

$$4.167(1-D) = 1+D$$
$$4.167 - 4.167D = 1+D$$
$$3.167 = 5.167D$$
$$D = 0.613$$

Therefore,

Duty Ratio, $D \approx 0.61$

2. Inductor Design

The inductance is calculated as

$$L = \frac{V_{in} \times D}{f_s \times \Delta I_L}$$

Where,

- $V_{in} = 48$ V
- $D = 0.61$
- Switching Frequency, $f_s = 50$ kHz
- Output Current, $I_o = 1$ A

Assuming the inductor current ripple,

$$\Delta I_L = 0.25 \times I_o$$
$$\Delta I_L = 0.25 \times 1 = 0.25 \text{ A}$$

Substituting,

$$L = \frac{48 \times 0.61}{50 \times 10^3 \times 0.25}$$
$$L = \frac{29.28}{12500}$$
$$L = 0.00234 \text{ H}$$
$$\boxed{L \approx 2.34 \text{ mH}}$$

3. Output Capacitor Design

The output capacitor is calculated using

$$C = \frac{I_o \times D}{f_s \times \Delta V_o}$$

Assuming an output voltage ripple of 5%,

$$\Delta V_o = 0.05 \times 200 = 10 \text{ V}$$

Substituting,

$$C = \frac{1 \times 0.61}{50 \times 10^3 \times 10}$$
$$C = \frac{0.61}{500000}$$
$$C = 1.22 \times 10^{-6} \text{ F}$$
$$\boxed{C \approx 1.22\ \mu\text{F}}$$

Final design parameters

In this project, inductors of 180 μH and an output capacitor of 22 μF were utilized due to component availability constraints. Despite these deviations from the calculated design values, the converter was successfully implemented in the Simulink environment, where the desired output voltage was achieved, demonstrating satisfactory performance of the system.